

Sustainability Dashboard User Guide

Purpose

This guide explains how we use the Sustainability Dashboard in the current SentinaAI build. It is written for sustainability managers and team members who need to monitor energy use, environmental conditions, hall level performance, alerts, devices, navigation views, digital twin access, and sustainability reporting.

The Sustainability Dashboard is built to help us track live energy trends, carbon estimates, HVAC efficiency, environmental health, hall level sustainability signals, and report generation in one connected workspace.

Who this guide is for

We use this guide when we need to:

- monitor sustainability performance across the venue
- review live energy and environmental trends
- inspect hall level sustainability details
- investigate alerts and exceptions
- check devices supporting sustainability readings
- use navigation and digital twin tools
- create, edit, generate, preview, download, or delete reports

Before we begin

Before using the Sustainability Dashboard, we should make sure:

- we are logged in with a Sustainability Manager account
- the backend services are running
- live analytics and AI services are available
- energy, environmental, and hall level data feeds are updating normally

If services are delayed or unavailable, some cards may show fallback values, empty states, or error messages.

Main layout

The Sustainability Dashboard uses a consistent layout, but with green themed indicators.

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Left sidebar

The left sidebar is the main way to move through the Sustainability section. In the current build, it includes:

- Dashboard
- Devices
- Alerts
- Energy
- Environmental
- Navigation
- Digital Twin
- Reports
- Settings
- Help
- Logout

The sidebar can be collapsed or expanded using the small chevron button near the top.

Top header

The top header includes:

- the page title
- a short subtitle showing that we are in the Sustainability Dashboard
- the signed in user card

The page title changes automatically based on the page we open.

Settings

Settings open as an overlay. In the current build, the Sustainability settings area lets us:

- change live refresh cadence
- choose the preferred report export format
- review signed in account details
- change password

Logout

Logout clears the active session and returns us to the login page.

Dashboard home

The Dashboard page is the main sustainability overview. It is the best place to start when we want a quick reading of current venue efficiency and environmental performance.

KPI cards

At the top of the page, we see four main cards:

- **Current Energy Usage**
- **Carbon Emission Estimate**
- **HVAC Efficiency**
- **Automation Status**

These cards give a quick snapshot of the latest interval and AI backed sustainability status.

Automation Status

The Automation Status card is clickable. Selecting it opens a larger AI live status modal so we can review hall level sustainability signals in more detail.

Electricity Consumption

This panel shows a live trend for electricity consumption over the latest six hour view. It is useful for spotting spikes, dips, and sustained high usage.

Carbon Forecast Snapshot

This panel shows the short term carbon trend. We use it to check whether current conditions are moving toward a higher emission period.

Top Halls by Energy Use

This panel highlights the halls with the heaviest energy demand. It helps us identify where attention may be needed first.

Environment Health Score

This card uses a gauge style view to show the current environment health score. It acts as a quick summary of how healthy the overall operating environment is.

AI Sustainability: Live Status

This panel shows a hall by hall AI summary with these columns:

- Hall
- HVAC Energy
- Carbon

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- Efficiency
- Status
- AI Action
- Anomaly

This is one of the most important panels in the Sustainability area because it connects operational conditions with AI interpretation.

Each row is clickable and opens the hall details page for that hall.

Hall details page

The Hall Details page opens when we click a hall from the AI Sustainability live table.

What this page shows

The page is built around a single hall and includes:

- hall name and sustainability status badge
- Energy Usage
- Carbon Emissions
- Occupancy
- AI Insight
- Hall Overview
- Environment Metrics

AI Insight section

This section shows:

- **AI Action**
- **Anomaly Detected**

AI Action may return items such as:

- Schedule Maintenance
- Optimize HVAC
- Reduce Load
- No action required

Hall Overview

The Hall Overview section includes:

- Hall ID
- Zone

- Role
- Capacity
- Current Occupancy
- Efficiency Score

Environment Metrics

The Environment Metrics table includes:

- Indoor Temperature
- Outdoor Temperature
- Humidity
- Comfort Index

Navigation

The page includes a back button at the top and another back button at the bottom so we can return quickly to the previous view.

Devices page

The Devices page is shared with the main dashboard layout and is used to inspect connected assets that feed the platform.

What we can do here

We can:

- search devices
- filter by zone
- filter by hall
- filter by type
- filter by status
- sort results
- clear filters
- change rows per page
- move between pages of results

Filters

The current filters are:

- Search here
- Zones
- Halls

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- Types
- Status
- Sort

The hall list depends on the selected zones. Once we choose one or more zones, the hall options update to only show halls inside those selected zones.

Table columns

The device table includes:

- Device ID
- Type
- Zone
- Hall
- Edge
- Status
- Last Communicated
- Metrics
- MAC

When to use this page

We usually open Devices when:

- a sustainability reading looks unusual and we want to trace the source
- we want to confirm whether a hall has the expected connected devices online
- we want to narrow issues by zone, hall, type, or status

Alerts page

The Alerts page is the working list for active and historical alerts, including alerts that may affect sustainability monitoring.

What we can do here

We can:

- search by message, rule, zone, hall, or device
- filter by severity
- filter by status
- filter by zone
- filter by hall
- filter by rule

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- sort results
- acknowledge new alerts
- resolve alerts that are still open
- open the full alert detail page

Filters

The current filters are:

- Search
- Severity
- Status
- Zone
- Hall
- Rule
- Sort

Table columns

The alerts table shows:

- Detected
- Severity
- Status
- Rule
- Zone
- Hall
- Device
- Trigger
- Recommended Action
- Actions

Alert actions

From the list page, we can use:

- **Ack** to acknowledge a new alert
- **Resolve** to mark the alert as resolved

Acknowledge is only available for new alerts. Resolve is disabled when an alert is already resolved or closed.

Alert details page

Clicking an alert row opens the full alert detail page. This view shows:

- Alert ID
- Root Cause
- Priority
- Zone
- Time
- Description
- Source
- AI Confidence
- Threshold
- Current Occupancy or Trigger Value
- Recommended actions

Where supported by the backend, actions can be selected and executed from this page.

Energy page

The Energy page is the focused workspace for electricity and energy performance.

What we can do here

We can:

- search energy rows
- filter by zone
- change the metric view
- change sort order
- download a report from this page

Filters

The current controls are:

- **Search here**
- **Zone**
- **Metrics**
- **Sort**
- **Download Report**

Metric options

The current metric selector includes:

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- Metrics: HVAC energy
- Metrics: Temperature
- Metrics: Carbon

Main panels

The Energy page includes:

- **Live Energy Usage**
- **Average Consumption Per Zone (kWh)**
- **Energy Consumption By Device Type**
- **Total Energy (24h)**
- **Peak Interval (24h)**
- **HVAC Share (24h)**
- **Anomalies / Alerts**

How to use this page

This page is useful when we need to answer questions such as:

- which zones are consuming the most energy
- whether HVAC is taking a large share of total energy use
- whether energy anomalies appeared in the last 24 hours
- whether usage is trending up or down

Environmental page

The Environmental page is the focused workspace for air quality, temperature, humidity, comfort, and carbon related environmental signals.

What we can do here

We can:

- search environment rows
- filter by zone
- switch between environmental metrics
- sort the current view
- download an environmental report from the page

Filters

The current controls are:

- Search here

- Zone
- Metrics
- Sort
- Download Report

Metric options

The current metric selector includes:

- Metrics: Air quality
- Metrics: Temperature
- Metrics: Humidity
- Metrics: Carbon
- Metrics: Efficiency
- Metrics: Comfort

Main panels

The Environmental page includes:

- **Live Environmental Signal**
- **Average by Zone (24h)**
- **Anomalies / Alerts**
- **Total Carbon (24h)**
- **Avg Indoor Temp (24h)**
- **Avg Humidity (24h)**
- **Air Quality Score (24h)**

What this page is good for

We use it when we need to understand whether conditions are comfortable, stable, and environmentally efficient across zones.

It is especially useful when we want to compare comfort and environmental quality alongside carbon and humidity trends.

Navigation page

The Navigation page is used for route finding inside the venue.

What we can do here

We can:

- choose a start location

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- choose an end location
- click **Find Path**
- switch **Heatmap** on or off
- switch **Avoid Crowds** on or off
- switch **Demo IoT** on or off
- review the route on the map

How to use it

1. Select the start point.
2. Select the end point.
3. Keep **Avoid Crowds** on if we want the route to favour lower crowd pressure.
4. Click **Find Path**.
5. Review the route shown on the map.

The text below the map shows either the number of route waypoints or a prompt to choose start and end points first.

Digital Twin page

The Digital Twin page opens the digital twin experience inside the main dashboard shell.

What this page is for

We use the Digital Twin when we want a spatial and visual view of the venue rather than only charts and tables.

In the current build, this page is embedded through an iframe, so the exact tools available inside it depend on the digital twin service that is running.

Reports page

The Reports page is where we manage saved and generated sustainability reports.

What we can do here

We can:

- search reports
- filter by report type
- filter by format
- filter by status
- filter by date
- sort reports
- create a new report

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- generate a saved draft
- edit a draft
- preview a generated PDF
- download generated files
- delete a draft or generated report

Filters

The current filters are:

- Search here
- Report Type
- date filter where available
- Format
- Status
- Sort

Actions by report status

For **Draft** reports, the available actions are:

- Generate draft
- Edit draft
- Delete draft

For generated **PDF** reports, the available actions are:

- Download
- Preview
- Delete

For generated **XLSX** reports, the available actions are:

- Download
- Delete

Why this matters

This page lets us separate report preparation from final output. We can save work in progress first, then generate the final file once the filters and sections are correct.

Generate Report page

The Generate Report page is where we create a new sustainability report or edit an existing draft. Data range is 24/12/2025 to 25/01/2026.

Report configuration

The page includes:

- report title
- date from
- date to
- zones
- halls
- included sections
- frequency
- custom notes
- report format

Current sustainability report sections

In the current build, the Sustainability report can include:

- Energy Consumption
- Environmental Conditions
- Occupancy Overview

Filter behaviour

Zone selection controls the hall list. Once zones are selected, the hall list updates to only show halls from those zones.

Frequency options

The page includes these options:

- Hourly
- Daily
- Weekly
- Monthly

File format

The current page allows us to choose:

- PDF
- XLSX

Main actions

At the bottom of the page, we can use:

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- **Cancel** to leave without saving
- **Save Draft** to keep the current configuration for later
- **Generate** to create the final report output

Recommended way to use this page

1. Enter a clear report title.
2. Set the reporting date range.
3. Select the relevant zones.
4. Select one or more halls if needed.
5. Keep or adjust the included sections.
6. Choose the frequency.
7. Add notes if context is needed.
8. Choose PDF or XLSX.
9. Save as draft if the report needs review.
10. Generate once the setup is final.

Search, filters, and paging

Most list pages in the Sustainability section follow the same working pattern:

1. Use the search box inside the page controls if we know the item name or ID.
2. Apply filters to narrow the list.
3. Use sort only after the list is already focused.
4. Move between pages where paging is available.
5. Clear filters when starting a fresh search.

This keeps the workflow clean and reduces the chance of carrying over an old filter.

Good working practice

To use the Sustainability Dashboard well, we should:

- start with the Dashboard page for the overall picture
- open the AI Sustainability live table when hall level performance needs closer review
- move into Hall Details for a specific hall when the AI table shows a concern
- use Energy when the issue is mainly energy related
- use Environmental when the issue is mainly comfort, humidity, air quality, or carbon related
- use Alerts and Devices when the signal may be caused by a rule trigger or device issue
- use Navigation and Digital Twin when a spatial view is needed
- use Reports when a file output or management summary is required

Common issues

No data on a page

Check whether:

- the backend service is running
- the correct environment URL is set
- the live AI endpoints are available
- the selected filters are too narrow
- the chosen date range excludes the needed records

No halls appear after choosing a zone

This usually means:

- the selected zone has no mapped halls in the current result
- the data source did not return halls for that zone
- the zone filter needs to be cleared and selected again

Buttons are disabled on the Alerts page

This is normal in some cases:

- **Ack** is disabled if the alert is not new
- **Resolve** is disabled if the alert is already resolved or closed

Report actions look different between rows

This is also normal. Report actions change based on status and file type.

Rule-Based Assistant

The Rule-Based Assistant is available throughout the Sustainability Dashboard as a floating button in the bottom right corner of the screen. It can be opened from any Sustainability page without leaving the current view.

When opened, the assistant appears as a panel titled **Senti Sustainability**. The panel includes the same assistant controls used in the other dashboards, including expand, reset, and close.

At the start of a session, the assistant shows a **Current analysis** chip and a guided welcome message. It asks the user to use a structured workflow to analyse sustainability by zone, hall, and date range.

The assistant offers the following starting options:

- Overview

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- Energy
- Comfort
- By Event Overview
- Efficiency & Carbon

This allows the user to begin with the exact sustainability topic they want to review.

How the assistant is used

The Sustainability assistant follows the same guided workflow approach. Once a topic is selected, the assistant leads the user through the analysis setup instead of requiring them to build the view manually.

The assistant is designed to help users work through sustainability questions in a structured way, especially when reviewing:

- energy use
- comfort conditions
- event-based sustainability performance
- efficiency and carbon patterns

Quick actions and working flow

The assistant includes a **Quick actions** section at the bottom of the panel.

This keeps the sustainability workflow consistent and gives users a simple path to continue, restart, or move to supporting guidance after an analysis is completed.

What this feature is for

The Rule-Based Assistant is designed to make sustainability analysis easier to follow and easier to repeat. It helps users move from a sustainability question to a focused result in a guided way.

It is especially useful when the user wants to:

- review sustainability performance quickly
- focus on one topic at a time
- analyse a zone, hall, or date range without manually building the full view
- keep the workflow simple during live review

Current build note

This guide reflects the Sustainability pages currently wired into the main dashboard build:

- Dashboard
- Hall Details
- Devices
- Alerts
- Alert Details
- Energy
- Environmental
- Navigation
- Digital Twin
- Reports
- Rule-based assistant
- Generate Report
- Settings overlay